# Abdul Rahman Shaikh

+1-224-848-2725 iamsabdurahman@gmail.com Chicago, IL sabdulrahman.github.io
linkedin.com/in/iamsabdurahman github.com/sabdulrahman Google Scholar

## Area of Expertise

Ph.D. candidate in Computer Science specializing in LLMs, multimodal learning, and large-scale AI systems. I design and deploy retrieval pipelines, multi-agent workflows, and evaluation frameworks for high-volume unstructured data. My work focuses on building scalable, production-ready systems that improve analysis speed and decision-making. I have led research projects, mentored student researchers, and published 10+ papers in venues such as IEEE VIS and TVCG.

## Education

**Northern Illinois University** — Dekalb, IL
*Ph.D. Computer Science (GPA: 3.9)* — *Aug 2020 – Present*

**Northern Illinois University** — Dekalb, IL
*M.S. Computer Science (GPA: 3.9)* — *Jan 2018 – May 2020*
*Thesis: Modeling the Broader Impact of Science and Health Using Social Media*

**Osmania University** — Hyderabad, India
*B.E. Computer Science (GPA: 3.6)* — *Aug 2013 – Aug 2017*

## Experience

**AI/ML Engineer** — Remote
*Visual Technologies* — *2024 – Present*

- Built and deployed production RAG pipelines to process large-scale unstructured data (customer feedback, reports), reducing insight extraction time from days to minutes.
- Fine-tuned LLMs (LLaMA, Mixtral) using LoRA/QLoRA, reducing GPU memory usage by 60% and training time by 45% while maintaining performance.
- Optimized inference with quantization (GPTQ, AWQ) and distillation, achieving 4× throughput for real-time applications.
- Developed automated evaluation pipelines (Ragas, TruLens, LLM-as-judge) to measure retrieval accuracy, hallucination, and bias across deployments.
- Architected multi-agent workflows (LangChain, LangGraph, AutoGen) to automate retrieval and synthesis, reducing analysis turnaround time from weeks to days.
- Built containerized inference services (FastAPI, Docker, Kubernetes) with CI/CD pipelines, enabling scalable and reliable deployment of LLM systems in production environments.

**AI/ML Researcher (DATA Lab, VA Lab & WASTE Lab)** — Dekalb, IL
*Northern Illinois University* — *Jan 2018 – Present*

- Built LLM and ML pipelines processing 1M+ multimodal records, enabling large-scale experimentation and downstream analytics across 3 research labs.
- Led the Visual Analytics Lab, mentoring 5+ researchers and delivering 4 publications and multiple deployed systems.
- Architected LLM-powered analytical pipelines for reasoning, retrieval, and multimodal analysis, improving task accuracy and consistency through evaluation-driven iteration.
- Designed evaluation frameworks for LLM and ML systems (benchmarking, ablation studies, error analysis), improving reproducibility and consistency across experiments.
- Developed multi-agent workflows (LangChain, AutoGen, CrewAI) for automated retrieval and synthesis, reducing literature review time by 50%+.
- Built computer vision pipelines (ViT, SAM, diffusion augmentation) for classification and segmentation across heterogeneous datasets.
- Published 10+ papers in IEEE VIS, TVCG, and JCDL on data analysis and AI systems.

**Teaching Assistant (Algorithms, Databases, C/C++, & JAVA)** Dekalb, IL

*Northern Illinois University* *Aug 2018 – May 2023*

- Provided academic support to over 70 students, facilitating a deeper understanding of algorithms, C/C++ programming concepts, memory management, data structures, and SQL database management.
- Evaluated assignments emphasizing efficient algorithm design, optimized database queries, and real-world problem-solving in C++ and SQL, ensuring alignment with industry standards.
- Conducted weekly sessions and personalized mentoring, improving students' problem-solving abilities and leading to a 20% increase in class performance metrics.

**Data Analyst** Hyderabad, India

*Amazon* *Aug 2016 – Dec 2017*

- Implemented SQL and MongoDB pipelines to optimize data querying for performance bottlenecks, reducing database query latency by 30%.
- Utilized Python libraries (Pandas, NumPy, Scikit-learn) to perform detailed data cleansing, exploratory analysis, and quality assurance, significantly enhancing the reliability of critical datasets.
- Collaborated with cross-functional development teams to provide actionable, data-driven insights that improved database system performance and streamlined operations.

## Publications

### Published

- **A. R. Shaikh**, M. Sun, X. Liu, H. Alhoori, J. Zhao, and D. Koop, "iTrace: Interactive Tracing of Cross-View Data Relationships", *Graphics Interface 2025.* [doi: 10.48550/arXiv.2505.23079]
- **A. R. Shaikh**, H. Alhoori, and M. Sun, "YouTube and Science: Models for Research Impact," *Journal of Scientometrics*, 2022. [doi: 10.1007/s11192-022-04574-5]
- **A. R. Shaikh**, M. Sun, and H. Alhoori, "Toward systematic design considerations of organizing multiple views", *IEEE Visualization and Visual Analytics (VIS)*, 2022. [doi: 10.1109/VIS54862.2022.00030]
- M. Rezaei, M. R. Moghadam, **A. R. Shaikh**, H. Alhoori, and R. Freedman, "Generation, Evaluation, and Explanation of Novelists' Styles with Single-Token Prompts," *ACM/IEEE Joint Conference on Digital Libraries, JCDL, Dec 15–19, 2025.* [doi:10.48550/arXiv.2511.20459]
- M. J. Mokarrama, **A. R. Shaikh**, and H. Alhoori, "Examining the Representation of Youth in the US Policy Documents through the Lens of Research," in *2024 IEEE International Conference on Big Data*, Washington, USA, Dec 15–18, 2024. [doi: 10.1109/BigData62323.2024.10825996]
- M. Sun, **A. R. Shaikh**, Y. Ma, D. Koop, and H. Alhoori, "Boundary Blending: Reconsidering the Design of Multi-View Visualizations," 2023. [doi: 10.48550/arXiv.2306.09812]
- M. Sun, **A. R. Shaikh**, H. Alhoori, and J. Zhao, "SightBi: Exploring Cross-View Data Relationships with Biclusters," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 1, 2022. (Proceedings of IEEE VIS 2021). ***Best Paper Honorable Mention.*** [doi: 10.1109/TVCG.2021.3114801]
- M. Shahzad, H. Alhoori, Reva Freedman, and **A. R. Shaikh**, "Quantifying the online long-term interest in research," *Journal of Informetrics*, 2022. [doi: 10.1016/j.joi.2022.101288]
- **A. R. Shaikh** and H. Alhoori, "Predicting Patent Citations to measure Economic Impact of Scholarly Research," in *Proceedings of ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, Champaign, Illinois, USA, June 2–6, 2019. [doi: 10.1109/JCDL.2019.00089]
- **A. R. Shaikh**, "Modeling the Broader Impact of Science and Health Using Social Media," Master's thesis. [link]

### Under Review

- M. Sun, **A. R. Shaikh**, Y. Ma, D. Koop, and H. Alhoori, "Toward the Design of Transformative Multiple-View Visualizations". *PacificVis2026*
- N. Vooda, M. Uddin, **A. R. Shaikh**, M. Karasani, M. Hughes and M. Vaezi, "USS-Water Dataset and U-Net+ Model: A Novel High-Resolution Satellite Imagery Approach for Surface Water Detection in the United States". *PLOS Water Journal*
- **A. R. Shaikh**, M. Ambati, M. Uddin, and M. Vaezi, "Building Classification: A Comprehensive Dataset and DenseNet201-Based Approach". *Journal of Urban Planning and Development*
- M. Karasani, **A. R. Shaikh**, M. Uddin, and M. Vaezi, "Machine-Learning-Based County-Level Prediction of Municipal Solid Waste in the United States with Socioeconomic and Demographic Integration". *Waste Management & Research: The Journal for a Sustainable Circular Economy (WM&R)*

## Projects

**GenHealth: Multimodal Medical Report Analysis** Code | *Transformers, FastAPI* May 2025 – July 2025

- GenHealth is a multimodal AI system that combines text processing, medical imaging analysis, and structured data integration to boost diagnostic extraction accuracy from medical reports.

**Rufus: Intelligent Web Data Extraction for LLMs** Code | *RAG, Selenium* Jan 2025 – Mar 2025

- Developed an AI-powered web crawler that extracts and synthesizes relevant content into structured documents for Retrieval-Augmented Generation (RAG) systems.

**LLMFlow: AI-Powered Summarization of Scholarly Documents** Code | *LangChain* Jun 2024 – Dec 2024

- Integrated GPT, Claude and Llama with LangChain to generate concise summaries of large-scale research documents, reducing academic reading time.

**MixArt: Generative Artwork with Stable Diffusion** Code | *Stable Diffusion, LoRA* Jan 2024 – July 2024

- Developed a pipeline for generating abstract and photorealistic art using Stable Diffusion, leveraging LoRA fine-tuning to personalize outputs for user-requested styles.

**VoxCore: Voice Authentication System** Code | *Python, Whisper, PyTorch* June 2023 – Sep 2023

- Implemented a voice authentication system using OpenAI's Whisper for transcription and custom PyTorch models for speaker verification, achieving 89% accuracy in multi-speaker environments.

**InstaHealth: Fine-Tuning Caption Generation for Instagram Posts** Code | *CV* Aug. 2022 – Dec 2022

- Fine-tuned a language model to generate context-aware captions for health-related content on Instagram data, automating large-scale social media captioning.

**Semantic Segmentation with SAM & Custom U-Net** | *Python, CV* Jan 2021 – May 2021

- Integrated the Segment Anything Model (SAM) with a custom U-Net architecture to achieve semantic segmentation on high-resolution aerial images.

**TweeTopics: Uncovering Patterns with BERT and LDA** | *Python, BERT, LDA* Aug. 2020 – Dec. 2020

- Implemented BERT and LDA for deep learning text and topic modeling, employing unsupervised learning algorithms to analyze social discourse dynamics on Twitter.

**Urban Pedals: Visualizing BikeShare Dynamics in Chicago** | *Python, R, Tableau* Jan. 2019 – July 2019

- Utilized Pandas, R for data analysis, and Tableau, JavaScript for visualizations, applying clustering algorithms and time-series analysis to uncover patterns in Chicago's BikeShare system, revealing urban transportation patterns.

**Predicting popularity using Altmetric** | *Python, ScikitLearn, API (Altmetric)* Jan. 2018 – May 2018

- Employed ScikitLearn and Python for predictive modeling using regression algorithms and decision trees to forecast future citation counts of scholarly articles using Altmetric data, identifying early impactful research.

## Invited Talks & Presentations

- **Fluid Multi-View Analytics: Generative Multi View Systems**, *DePaul University CDM*, Chicago, 2025.
- **iTrace: Interactive Tracing of Cross-View Data Relationships**, *University of British Columbia*, BC, 2025.
- **MSW Predictive Modeling: A comprehensive approach**, *CEET Innovation Showcase*, DeKalb, 2024
- **Multi-Class Building Classification**, *IIN Sustainability Research Conference*, Chicago, 2024
- **Utilizing Multiple Views to analyze data**, *Northern Illinois University, CSCI 658*, DeKalb, 2023.
- **Toward Systematic Design Considerations of Organizing Multiple Views**, *IEEE VIS* conference, 2022
- **Exploring COVID data using Instagram Images**, *Northern Illinois University, CSCI 656*, DeKalb, 2022.
- **Modeling the Broader Impact of Science**, *Northern Illinois University, CSCI 600*, DeKalb, 2022.
- **Organizing layout of Multiple View systems**, *Northern Illinois University, CSCI 628*, DeKalb, 2021.
- **Cross-Device interaction to explore 3D Vis**, *Northern Illinois University, CSCI 626*, DeKalb, 2021.
- **Predicting Patent Citations to measure Economic Impact**, *JCDL 2019*, Urbana-Champaign, 2019.
- **Visualizing BikeShare Dynamics in Chicago**, *Northern Illinois University, CSCI 627*, DeKalb, 2019.

## Honors & Awards

- **Best Paper Honorable Mention**, *IEEE VIS*, 2022
- **Society Involvement Award**, *Honor Society*, 2020
- **Community Service Award**, DeKalb, 2019

## Leadership

- **Lab Head**, *Visual Analytics Lab*, NIU, DeKalb, 2020 – present
- **Mentor**, *WASTE Lab*, NIU, DeKalb, 2024 – 2025
- **Chief Coordinator**, *Computer Society of MJCET*, Hyderabad, 2017
- **Leader – Web/Graphic Design Head**, *Entrepreneurship Cell, MJCET*, 2016
- **Lead Developer**, College's Annual Technical Fest Website (Adsophos), 2015

## Services

### Program Committee Member

- CIKM 2025
- WWW 2026
- JCDL 2025
- CIKM 2024
- CIKM 2023

### Reviewer

- Scientometrics Journal
- Frontier of Psychology
- New Media & Society
- PacificVis
- WebSci
- PLOS ONE

### Professional Memberships

- IEEE
- ACM

## Skills

**Programming:** Python, Java, C/C++, JavaScript/TypeScript, SQL, R, HTML/CSS, LaTeX
**ML & CV:** PyTorch, TensorFlow/Keras, Hugging Face, Scikit-learn, XGBoost/LightGBM, OpenCV, ViT, CLIP, SAM
**GenAI & LLM:** RAG, prompt engineering, tool/function calling, embeddings, LoRA/QLoRA (PEFT), quantization; OpenAI/Anthropic/Google APIs; LLaMA, Mixtral, Vicuna; Whisper; Stable Diffusion/DreamBooth
**Agents & Orchestration:** LangChain, LangGraph, AutoGen, CrewAI, ReAct, n8n, Flowise, LlamaIndex
**Retrieval & Data:** FAISS, ANN indexing, hybrid search (BM25+dense), reranking; PostgreSQL, MongoDB, Redis, Snowflake, BigQuery, Kafka
**Evaluation & Observability:** A/B testing, SHAP/LIME; hallucination/bias testing, golden sets; Ragas/TruLens/DeepEval, LangSmith
**MLOps & Deployment:** Docker, Kubernetes, FastAPI, MLflow, Weights & Biases, DVC, Git/GitHub Actions, CI/CD, model versioning, inference optimization
**Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit/Dash, Tableau/Power BI, D3.js